Property Rights: Public Goods and Common Pool Resources

Reference: Chapter 11, Mankiw

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February 12th, 2026

What are the factors

- Two key factors in property rights
  - Rivalrous: does one person's use of the good reduce another person's ability to use it?
  - Excludable: Can people be altogether prevented from using the good
  - As you can imagine, there are 4 possibilities.

Different Kinds of Goods

Excludable?	Rival in consumption? Yes	Rival in consumption? No
Yes	Private Goods • Ice-cream cones • Clothing • Congested toll roads	Club Goods • Fire protection • Cable TV • Uncongested toll roads
No	Common Resources • Fish in the ocean • The environment • Congested nontoll roads	Public Goods • Tornado siren • National defense • Uncongested nontoll roads

FIGURE 1

Four Types of Goods

Goods can be grouped into four categories according to two characteristics: (1) A good is *excludable* if people can be prevented from using it. (2) A good is *rival in consumption* if one person's use of the good diminishes other people's use of it. This diagram gives examples of goods in each category.

Public goods

- This is not as straightforward as one may think.
- Of course a simple example is air.
- But for example, with something like national defense, it is largely a public good, but someone does pay for it.
- However, it is still non-rivalrous and non-excludable.

A typical public goods problem from game theory

Group Project. A group project consists of $i \in \{1, 2\}$ students. Each student exerts an effort level e_i, and the aggregate effort level is given by

$$E = e_1 + e_2.$$

A student's grade function g_i is given by

$$g_i(e_i, e_j) = (t_i - e_i)\sqrt{mE}$$

where t_i is the total time available to player i and m (marks) is the aggregate returns on effort.

Nature of the above problem

- One can potentially increase their utility by putting effort here (unlike a good like air).
- However, if they did nothing and the other person does something, they enjoy some marks for free.
- This concept is known as "free-riding".
- Incentive to free ride leads to several sup-optimal outcomes.

Role of government

- Even in the most conservative/capitalist economies, there is a consensus that government is needed to invest in public goods.
- These can be roads, hospitals, water supply etc.
- The idea is to do some sort of cost-benefit analysis.
- However, this analysis cannot simply be a revenue-cost comparison.

Common Pool Resources

- There are certain resources that people cannot be excluded from using, but are rivalrous.
- this means that when one person uses the good, it makes it more difficult for another to do the same.
- We can think of it in two ways:
  - When one person uses the good (a resource), future usage becomes more costly (fisheries).
  - When one person is using that resource, others have to wait (a fully congested road)

How a problem looks

Cost externalities. Two firms ($N = 2$) exploit a common resource stock S. Each firm $i \in \{1, 2\}$ chooses appropriation $q_i \geq 0$. Firms take prices as given (normalized to 1), and $\theta \in [0, 1]$ denotes severity of cost externality. Total cost of production of q_i units is given by

$$C_i(q_i, q_j) = \frac{q_i(q_i + \theta q_j)}{S}, \quad \theta \in [0, 1].$$

What is the issue above?

- There is a slightly different problem here.
- Consumption makes costs high for another person, leading to over-consumption of the resource.
- This is known as the "Tragedy of the Commons".
- There is also an interesting concept know as the Tragedy of the Anti-Commons.